

Haematological Malignancies

Lymphomas, Leukaemias, and Myeloma — Diagnosis and Treatment

Doc #10 of 15	Domain: Oncology & Cancer Treatment	Audience: Medical Professionals
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Keywords: lymphoma | leukaemia | myeloma | AML | CLL | DLBCL | CAR-T | bortezomib | venetoclax

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1. Classification Overview

Haematological malignancies are classified by WHO 2022 classification based on lineage (lymphoid vs. myeloid), differentiation stage (mature vs. precursor), and molecular/genetic features. Key changes in WHO 2022 include separation of AML with mutated NPM1 as a distinct entity, refined MDS categories, and incorporation of molecular data into lymphoma subtypes.

2. Diffuse Large B-Cell Lymphoma (DLBCL)

- Most common NHL (~30% of B-cell lymphomas); aggressive but potentially curable
- Cell-of-origin (COO): Germinal centre B-cell (GCB) vs. Activated B-cell (ABC)/non-GCB by IHC (Hans algorithm) or gene expression profiling (Lymph2Cx)
- Double-hit (DH) and Triple-hit: Concurrent MYC + BCL2 and/or BCL6 rearrangements by FISH; confers poor prognosis; treat with DA-EPOCH-R
- Standard: R-CHOP x6 cycles; PET-CT at end of treatment; CR rate ~70-80%
- Relapsed/refractory: CAR-T cell therapy (axicabtagene, lisocabtagene, tisagenlecleucel) for chemo-refractory; auto-SCT if chemo-sensitive; loncastuximab or polatuzumab vedotin-based regimens
- Bispecific T-cell engagers (glofitamab, epcoritamab) — emerging options in R/R setting with high ORR

3. Acute Myeloid Leukaemia (AML) — Molecular Classification

Molecular Group	Example Alteration	ELN 2022 Risk	Targeted Option
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Favourable	t(8;21), inv(16), NPM1-mut (FLT3 ITD-neg)	Favourable	Standard 7+3; gemtuzumab ozogamicin
Intermediate	NPM1-mut + FLT3 ITD, t(9;11)	Intermediate	7+3; consider allo-SCT in 1CR
Adverse	TP53-mut, RUNX1-mut, FLT3 ITD-hi, -5/-7	Adverse	Venetoclax + azacitidine (unfit); allo-SCT if fit
FLT3-mut (ITD or TKD)	FLT3 ITD/D835	Variable	Midostaurin (1L); Gilteritinib (R/R)
IDH1-mut	R132H	Intermediate/Adverse	Ivosidenib (1L unfit or R/R)
IDH2-mut	R140Q, R172K	Intermediate	Enasidenib (R/R)

4. Chronic Lymphocytic Leukaemia (CLL)

- Most common adult leukaemia in the West; indolent but incurable without SCT
- Key prognostic markers: del(17p)/TP53 mutation (worst prognosis), del(11q), unmutated IGHV, complex karyotype, ZAP-70, CD38
- First-line: Ibrutinib (BTK inhibitor) or Acalabrutinib ± obinutuzumab for high-risk; venetoclax + obinutuzumab (time-limited) for fit patients — CLL14 trial
- Richter transformation to DLBCL in ~5-10%: poor prognosis; consider platinum + anti-CD20 then CAR-T or allo-SCT

5. Multiple Myeloma (MM)

Disease Status	Standard Regimen	Notes
Newly diagnosed, SCT-eligible	VRd (bortezomib + lenalidomide + dex) induction → ASCT → lenalidomide maintenance	RVd superior to Rd (SWOG S0777); daratumumab addition investigated
Newly diagnosed, SCT-ineligible	Dara-VRd or Dara-Rd	MAIA trial: Dara-Rd superior OS vs Rd
Relapsed/Refractory	Carfilzomib + pom + dex; elotuzumab; selinexor	Belantamab mafodotin (BCMA ADC)
Penta-refractory	Idecabtagene vicleucel or ciltacabtagene (BCMA CAR-T)	KarMMa / CARTITUDE trials — high ORR ~70-98%

■ **TP53 / del(17p): Patients with TP53 mutation or del(17p) should receive BTK inhibitor-based or venetoclax-based regimens — they derive minimal benefit from standard chemoimmunotherapy.**